MILITARY SPECIFICATION

FORCEPS, BONE HOLDING, (Self-Centering)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bone holding forceps of small, medium and large sizes.

1.2 Classification. Forceps shall be of the following types as indicated below (see 6.2):

Type		
Small	-	9-7/16 inches (23.97 cm) long: 9/16 inch (14 mm) Jaws.
Medium	-	10-1/4 inches (26.04 cm) long: 23/32 inch (18 mm) Jaws.
Large	-	11-1/16 inches (28.10 cm) long: 29/32 inch (23 mm) Jaws.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

MSC N/A FSC 6515

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

2.1.2 Drawing. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWING

Defense Personnel Support Center (DPSC)

22326	- FORCEPS, BONE HOLDING

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(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General requirements. The bone holding forceps shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

3.2 Material. The arms of the forceps including the pin or screw that holds them together shall be fabricated from a martensitic 400 series corrosion-resisting steel. The locking mechanism, including the knurled nut, shall be fabricated from an austenitic 300 series corrosion-resisting steel. The corrosion-resisting steel shall be in accordance with ASTM F899.

3.3 Hardness. The Rockwell hardness of 420 stainless steel shall be Rc 44 to Rc 52. The Rockwell hardness of 302 stainless steel shall be Rb 80 to Rb 95.

3.4 Style, design and dimensions. The bone holding forceps shall be self-centering and suitable for use during orthopedic surgical cases. The forceps shall be in strict accordance with Defense Personnel Support Center (DPSC) Drawing #22326 for the appropriate type and National Stock Number (NSN).

3.5 Finish. Finish shall be satin or matte. The finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with fine abrasives under pressure and electropolish.

3.6. Passivation. The forceps shall be properly passivated.

3.7 Corrosion-resistance. Forceps shall be subjected to the boil test and copper sulfate test as specified in GG-I-526.

3.8 Identification markings. In addition to the requirements specified in GG-I-526, forceps of foreign manufacture shall be marked "stainless."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Inspection. Unless otherwise specified, inspection shall be in accordance with GG-I-526.

4.3 Quality conformance inspection. Certificates of quality, supplied by the manufacturer of the corrosion-resisting steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity, has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSNs, the item identification, the name of the material, the lot number, the lost size, the sample size, the date of testing, the test method, individual test results, and the the specification requirements.

4.3.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.4 Sampling.

4.4.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purpose shall be one forceps.

TABLE I. Sampling criteria for examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.4.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purpose shall be one ear forceps. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests

Characteristic	Inspection level	AQL (percent defective)
Hardness test	S-2	1.5
Dimensions	S-2	2.5
Copper sulfate test	S-2	1.0
Boil test	S-2	1.0

4.5. Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the forceps shall be selected for chemical analysis.

4.6 Examination.

4.6.1 Examination of forceps. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any forceps in the sample containing one or more defects shall be rejected and if the number of defective forceps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Forceps not of the specified design for appropriate type, stock number and item identification.
102	Forceps not free from cracks, pits, burrs, sharp edges, scale, rough areas or corrosion.
Minor	
201	Forceps not free from dirt, polishing material or foreign matter.
202	Identification markings not present, not correct, not complete, or not legible.

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4.7 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.7.1 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.8 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Forceps shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each forceps shall be packaged in accordance with MIL-P-116, Method IC-2.

5.1.2 Level C. The level C preservation for forceps shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twenty-four unit packages containing forceps of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

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5.2.2 Level B. Twenty-four unit packages containing forceps of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636 class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing for forceps shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.8.

6. NOTES

6.1 Intended use. The bone holding forceps is intended for use during orthopedic surgical procedures.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) National Stock Number.

6.3 National Stock Number coverage. This specification covers the following items:

National Stock No.	Item Identification	Specification Designation
6515-01-165-6744	FORCEPS, BONE HOLDING, Self-Centering, Curved, 10-1/4 Inches (26.04 cm) Long	Medium
6515-01-166-7824	FORCEPS, BONE HOLDING, Self-Centering, Curved, 9-7/16 inches (23.97 cm) Long	Small
6515-01-166-7825	FORCEPS, BONE HOLDING, Self-Centering, Curved, 11-1/16 inches (28.10 cm) Long	Large

6.4 Subject term (key word) listing.

Bone holding forceps
Curved forceps
Forceps
Self-centering forceps

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3499

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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